

PART A: TRUE / FALSE QUESTIONS**(10 MARKS)**

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| 1. A classification model can be used to predict the classification of an unknown entity. | TRUE / FALSE |
| 2. Ignoring instances that have missing values is one approach to dealing with missing values. | TRUE / FALSE |
| 3. Classification, Association Rule mining, Clustering and Prediction are typical data mining tasks. | TRUE / FALSE |
| 4. Dimensionality of data simplifies the analysis activities. | TRUE / FALSE |
| 5. In a decision tree, the root attributes has lower information gain among all of the other attributes. | TRUE / FALSE |
| 6. Supervised learning differs from unsupervised clustering in that supervised learning requires at least one output attribute. | TRUE / FALSE |
| 7. The K-Means clustering algorithm that we studied will automatically find the best value of k as part of its normal operation. | TRUE / FALSE |
| 8. A density-based clustering algorithm can generate non-globular clusters. | TRUE / FALSE |
| 9. In association rule mining the generation of the frequent item sets is the computational intensive step. | TRUE / FALSE |
| 10. A critical aspects with K-Means is the selection of starting instances. | TRUE / FALSE |

PART B: STRUCTURED QUESTIONS

(90 MARKS)

QUESTION 1 - DATA MINING & DATA PREPROCESSING

(25 MARKS)

- a) Handling unstructured data is one of the main challenges in **Text Mining**. Describe in details the necessary steps that are needed to provide a **structured representation** of text documents. (9 Marks)
- b) Discuss **THREE (3)** reasons of why it is necessary to do feature selection. (6 Marks)
- c) Determine whether or not each of the following activities is a data mining task. Justify your answer.
- i. Dividing the customers of a company according to their profitability. (2 Marks)
 - ii. Predicting the future stock price of a company using historical records. (2 Marks)
 - iii. Monitoring and predicting failures in a hydropower plant. (2 Marks)
 - iv. Extracting the frequencies of a sound wave. (2 Marks)
 - v. Monitoring the heart rate of a patient for abnormalities. (2 Marks)

QUESTION 2 – CLASSIFICATION

(25 MARKS)

- a) For a two-class classification problem, with a Positive class P and a negative class N, we can describe the performance of the algorithm using the following terms: TP, FP, TN, and FN.
- i. What do each of these terms refer to? (4 Marks)
 - TP:
 - TN:
 - FP:
 - FN:

- ii. Place the 4 terms listed above in part a into the appropriate slots in the table below. (4 Marks)

Table 1 : Evaluation Measurement

		Predicted	
		Positive	Negative
Actual	Positive		
	Negative		

- iii. Provide the formula for accuracy in terms of TP, TN, FP, and FN. (6 Marks)

- iv. Provide the formula for precision and recall using TP, TN, FP, and FN. (4 Marks)

- Precision =
- Recall =

- b) What are the major differences among the **THREE (3)** of classification techniques:
- Support Vector Machine (SVM)
 - Neural Network (NN)
 - Bayesian Belief Network

(7 Marks)

QUESTION 3 – CLUSTERING

(20 Marks)

- a) Suppose that the data mining task is to cluster the following eight points, with (x, y) representing location, into three cluster;

A1(2,10); A2(2, 5); A3(8, 4); B1(5, 8); B2(7,5); B3(6,4); C1(1,2); C2(4,9).

The distance function is Euclidean distance. Suppose initially we assign A1, B1, and C1 as the center of each cluster, respectively. Use the k-means algorithm to show;

- the three cluster centers after the first round of execution.
- the final three clusters.

$$\text{Euclidean distance}(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

(8 Marks)

- b) Compare and contrast the following pairs of approaches:
- Partitioning approach vs. Hierarchical approach
 - K-means vs. K-medoids
 - Single linkage vs. Complete linkage
- (6 Marks)
- c) Clustering has been popularly recognized as an important data mining task with broad applications. Give **TWO (2)** examples for each of the following cases:
- An application that takes clustering as a major data mining function.
- (3 Marks)
- An application that takes clustering as a preprocessing tool for data preparation for other data mining tasks.
- (3 Marks)

QUESTION 4 - ASSOCIATION RULES

(20 Marks)

- a) What is association analysis? How is it different from the other data mining tasks (e.g. classification, clustering, etc.)
- (4 Marks)
- b) Based on Apriori algorithm and Table 1:

Table 1: Transaction Database

Transaction ID	Items
10	Chocolate, Salad
20	Salad, Chocolate
30	Meat
40	Dairy, Pasta, Salad
50	Nuts, Pasta
60	Pasta, Chocolate
70	Pasta, Chocolate
80	Dairy, Pasta
90	Nuts, Meat, Pasta
100	Chocolate, Pasta

- Show the steps taken to generate the candidate itemsets and frequent itemsets (minimum support = 20.0%).
- (8 Marks)
- Generate all the association rules that can be discovered from transactions (minimum support = 20.0% and minimum confidence = 50%).
- (8 Marks)